

Q1.

The line has Cartesian equations $x - p = \frac{y+2}{q} = 3 - z$ and a plane has equation $\mathbf{r} \cdot \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} = -3$

- (a) In the case where the plane fully contains the line, find the values of p and q . (3)
- (b) In the case where the line intersects the plane at a single point, find the range of values of p and q . (3)

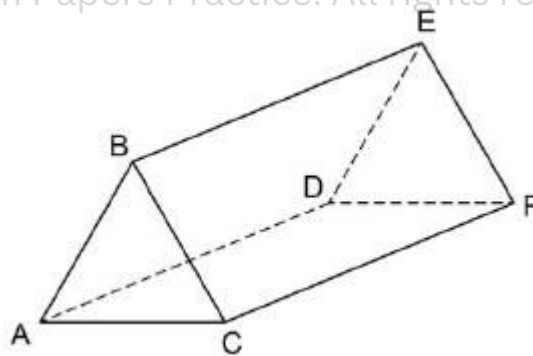
- (c) In the case where the angle θ between the line and the plane satisfies $\sin \theta = \frac{1}{\sqrt{6}}$ and the line intersects the plane at $z = 0$
- (i) Find the value of q . (4)
- (ii) Find the value of p . (3)

(Total 13 marks)

Q2.

A designer is using a computer aided design system to design part of a building. He models part of a roof as a triangular prism ABCDEF with parallel triangular ends ABC and DEF, and a rectangular base ACFD. He uses the metre as the unit of length.

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The coordinates of B, C and D are (3, 1, 11), (9, 3, 4) and (−4, 12, 4) respectively.

He uses the equation $x - 3y = 0$ for the plane ABC.

He uses $\left[\mathbf{r} - \begin{pmatrix} -4 \\ 12 \\ 4 \end{pmatrix} \right] \times \begin{pmatrix} 4 \\ -12 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ for the equation of the line AD.

Find the volume of the space enclosed inside this section of the roof.

(Total 9 marks)

Q3.

The four vertices of a parallelogram $ABCD$ have coordinates

$$A(1, 0, 2), B(3, -1, 5), C(7, 2, 4) \text{ and } D(5, 3, 1)$$

- (a) (i) Find $\overrightarrow{AB} \times \overrightarrow{AD}$. (3)

- (ii) Show that the area of the parallelogram is $p\sqrt{10}$, where p is an integer to be found. (2)

- (b) The diagonals AC and BD of the parallelogram meet at the point M . The line L passes through M and is perpendicular to the plane $ABCD$.

Find an equation for the line L , giving your answer in the form $(\mathbf{r} - \mathbf{u}) \times \mathbf{v} = \mathbf{0}$. (4)

- (c) The plane Π is parallel to the plane $ABCD$ and passes through the point $Q(6, 5, 17)$.

Find the coordinates of the point of intersection of the line L with the plane Π . (6)
(Total 15 marks)

Q4.

A line and a plane have equations

$$\frac{x-3}{p} = \frac{y-q}{3} = \frac{z-1}{-1}$$

and

$$\mathbf{r} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = 10$$

respectively, where p and q are constants.

- (a) Show that the line is **not** perpendicular to the plane. (1)

- (b) In the case where the line lies in the plane, find the values of p and q . (4)

- (c) In the case where the angle, θ , between the line and the plane satisfies $\sin \theta = \frac{1}{\sqrt{6}}$

, and the line intersects the plane at $z = 2$:

(i) find the value of p ;

(5)

(ii) find the value of q .

(2)

(Total 12 marks)

Q5.

The planes Π_1 and Π_2 have equations

$$\mathbf{r} \cdot \begin{bmatrix} 2 \\ 1 \\ 7 \end{bmatrix} = 10 \quad \text{and} \quad \mathbf{r} \cdot \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix} = 7$$

respectively.

(a) Determine, to the nearest degree, the acute angle between Π_1 and Π_2 .

(4)

(b) By setting $z = t$, find cartesian equations for the line of intersection of Π_1 and Π_2 in the form

$$\frac{x-a}{l} = \frac{y-b}{m} = z=t$$

(6)

(c) The line L , with equation $\mathbf{r} = \begin{bmatrix} 20 \\ -1 \\ 7 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 9 \\ 4 \end{bmatrix}$, intersects Π_1 at the point P and Π_2 at the point Q .

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Show that $PQ = k\sqrt{2}$, where k is an integer.

(6)

(Total 16 marks)

Q6.

The planes Π_1 , Π_2 and Π_3 have cartesian equations

$$2x + y - z = 3$$

$$3x - 2y + z = 5$$

$$12x - y - z = 40$$

respectively.

(a) Find, in the form $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$, a vector equation for the line L which is the intersection of Π_1 and Π_2 .

(5)

- (b) (i) Determine whether L meets Π_3 , and use your answer to decide whether the system given by the equations of these three planes is consistent or inconsistent.

(3)

- (ii) Describe geometrically the arrangement of the three planes.

(1)

- (c) (i) Find the coordinates of a common point of Π_2 and Π_3 .

(3)

- (ii) **Deduce** a vector equation for the line of intersection of Π_2 and Π_3 .

(1)

(Total 13 marks)

Q7.

The point Q has position vector $\mathbf{q} = \begin{bmatrix} 7 \\ 4 \\ 6 \end{bmatrix}$, the plane Π has equation $\mathbf{r} \cdot \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = 36$,
and the line l has equation $\mathbf{r} = \begin{bmatrix} 20 \\ -8 \\ 1 \end{bmatrix} + \mu \begin{bmatrix} -7 \\ 5 \\ 3 \end{bmatrix}$.

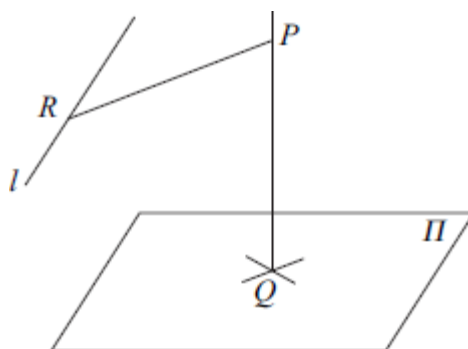
- (a) Show that Q lies in Π .

(1)

- (b) Show also that l is parallel to Π .

(2)

- (c) The diagram shows the point P , which lies on the normal to Π that passes through Q . The point R is the point on l which is closest to P , and $PQ = PR$.



Determine the coordinates of P .

(9)

(Total 12 marks)

Q8.

The matrices **A** and **B** are given in terms of p by

$$\mathbf{A} = \begin{bmatrix} 1 & p & 4 \\ -3 & 2 & 1 \\ 2 & -1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} p & 1 & 5 \\ 9 & p & -1 \\ 2 & 0 & 1 \end{bmatrix}$$

- (a) Find each of $\det \mathbf{A}$ and $\det \mathbf{B}$ in terms of p .

(3)

- (b) Without finding $\mathbf{AB}$, determine all values of p for which $\mathbf{AB}$ is singular.

(3)

(Total 6 marks)

Q9.

The line L and the plane Π have vector equations

$$\mathbf{r} = \begin{bmatrix} 7 \\ 8 \\ 50 \end{bmatrix} + t \begin{bmatrix} 6 \\ 2 \\ -9 \end{bmatrix} \quad \text{and} \quad \mathbf{r} = \begin{bmatrix} -2 \\ 0 \\ -25 \end{bmatrix} + \lambda \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix}$$

respectively.

- (a) Show that L is perpendicular to Π .

(3)

- (b) For the system of equations

$$\begin{aligned} 6p + 5q + r &= 9 \\ 2p + 3q + 6r &= 8 \\ -9p + 4q + 2r &= 75 \end{aligned}$$

form a pair of equations in p and q only, and hence find the unique solution of this system of equations.

(5)

- (c) It is given that L meets Π at the point P .

- (i) Demonstrate how the coordinates of P may be obtained from the system of equations in part (b).

(2)

- (ii) Hence determine the coordinates of P .

(2)

(Total 12 marks)

Q10.

Two fixed planes have equations

$$x - 2y + z = -1$$

- (a) The point P , whose z -coordinate is λ , lies on the line of intersection of these two planes. Find the x - and y -coordinates of P in terms of λ .

(3)

- (b) The point P also lies on the variable plane with equation $5x + ky + 17z = 1$. Show that

$$(k + 13)(2\lambda - 1) = 0$$

(3)

- (c) For the system of equations

$$\begin{aligned} x - 2y + z &= -1 \\ -x + y + 3z &= 3 \\ 5x + ky + 17z &= 1 \end{aligned}$$

determine the solution(s), if any, of the system, and their geometrical significance in relation to the three planes, in the cases:

- (i) $k = -13$;
 (ii) $k \neq -13$.

(6)

(Total 12 marks)

Q11.

Two planes have equations

$$\mathbf{r} \cdot \begin{bmatrix} 5 \\ 1 \\ -1 \end{bmatrix} = 12 \quad \text{and} \quad \mathbf{r} \cdot \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} = 7$$

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- (a) Find, to the nearest 0.1° , the acute angle between the two planes.
- (b) (i) The point $P(0, a, b)$ lies in both planes. Find the value of a and the value of b .
- (ii) By using a vector product, or otherwise, find a vector which is parallel to both planes.
- (iii) Find a vector equation for the line of intersection of the two planes.

(4)

(3)

(2)

(2)

(Total 11 marks)

Q12.

- (a) Find, to the nearest 0.1° , the acute angle between the planes with equations

$$\mathbf{r} \cdot (3\mathbf{i} - 4\mathbf{j} + \mathbf{k}) = 2 \quad \text{and} \quad \mathbf{r} \cdot (2\mathbf{i} + 12\mathbf{j} - \mathbf{k}) = 38$$

(4)

- (b) Write down cartesian equations for these two planes.

(2)

- (c) Find, in the form $\frac{x-a}{l} = \frac{y-b}{m} = \frac{z-c}{n}$, cartesian equations for the line of intersection of the two planes.

(5)

(Total 11 marks)

Q13.

The line l has equation $\mathbf{r} = \begin{bmatrix} 3 \\ 26 \\ -15 \end{bmatrix} + \lambda \begin{bmatrix} 8 \\ -4 \\ 1 \end{bmatrix}$.

- (a) Show that the point $P(-29, 42, -19)$ lies on l .
- (b) Find the acute angle between l and the z -axis.
- (c) The plane Π has cartesian equation $3x - 4y + 5z = 100$.
- (i) Write down a normal vector to Π .
- (ii) Find the acute angle between l and this normal vector.
- (d) Find the position vector of the point Q where l meets Π .
- (e) Determine the shortest distance from P to Π .

(1)

(1)

(1)

(4)

(4)

(3)

(Total 14 marks)

Q14.

(a) The plane Π has equation $\mathbf{r} = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} + \mu \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}$.

- (i) Find a vector which is perpendicular to both $\begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}$.

(2)

- (ii) Hence find an equation for Π in the form $\mathbf{r} \cdot \mathbf{n} = d$.

(2)

- (b) The line L has equation $\left(\mathbf{r} - \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix} \right) \times \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} = \mathbf{0}$.

Verify that $\mathbf{r} = \begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix} + t \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$ is also an equation for L .

(2)

- (c) Determine the position vector of the point of intersection of Π and L .

(4)

(Total 10 marks)

Q15.

Two planes, Π_1 and Π_2 , have equations $\mathbf{r} \cdot \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix} = 0$ and $\mathbf{r} \cdot \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix} = 0$ respectively.

- (a) Determine the cosine of the acute angle between Π_1 and Π_2 .

(4)

- (b) (i) Find $\begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix} \times \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$.

(2)

- (ii) Find a vector equation for the line of intersection of Π_1 and Π_2 .

(2)

(Total 8 marks)